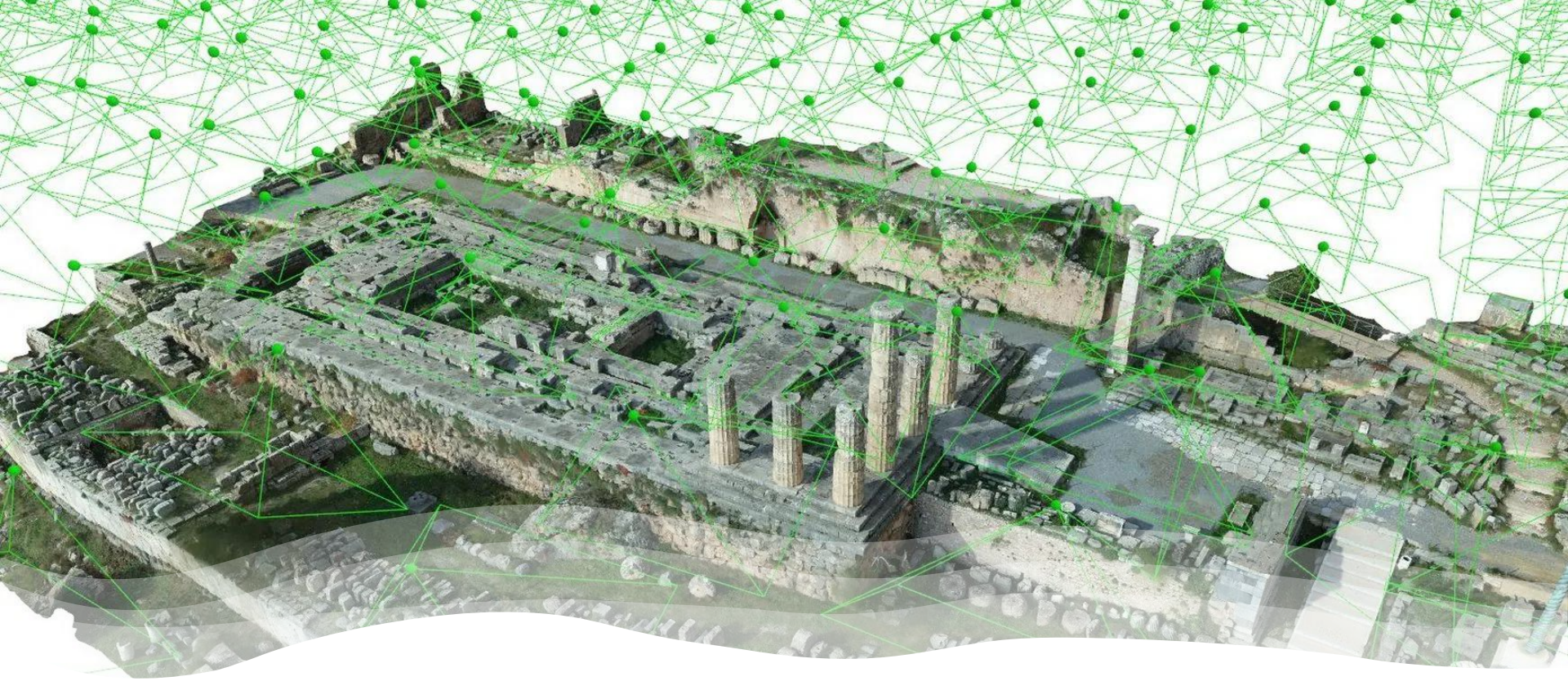


Advanced Topics in CAD

Photogrammetry for CAD

By: Jayanga Thanuka Samarasinghe, PhD Student, CIELO-G Fellow
University of Texas at El Paso

Outline

- What is Photogrammetry
- Applications with Examples
- How to do
- Our task

What is Photogrammetry ?

"Photogrammetry is the art, science, and technology of **obtaining reliable information** about **physical objects** and the **environment** through processes of recording, measuring, and interpreting **photographic images** and patterns of recorded radiant electromagnetic energy and other phenomena"

(Wolf and Dewitt, 2000)

Uses of Photogrammetry

- Topographic Mapping
- Develop Digital Surface Models (DSMs)
- Use as a tool for land surveys
- Spatial planning
- Applications in architecture, archeology, geomorphology, hydrology
- Traffic Management
- Medicine
- Military intelligence
- Disaster Response

Applications with Examples

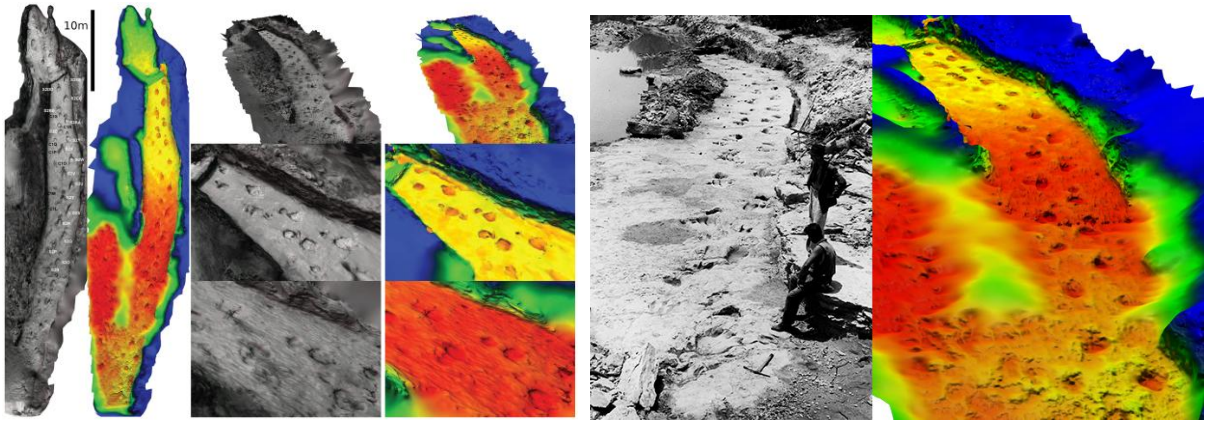
Famous Otzi the iceman: Source https://www.youtube.com/watch?v=Lr9jXJE363w

Actual

3D Model

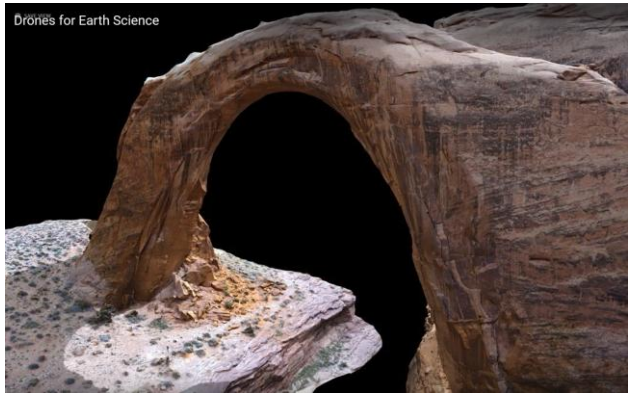
Rankoth Wehera, Sri Lanka
Source: Egodawela et al., 2022

Applications with Examples



Paluxy River dinosaur
'chase sequence'

Source: Falkingham et al., 2014



Source: Moore et al., 2016 and
https://www.youtube.com/watch?v=EVfCgDBwtFs

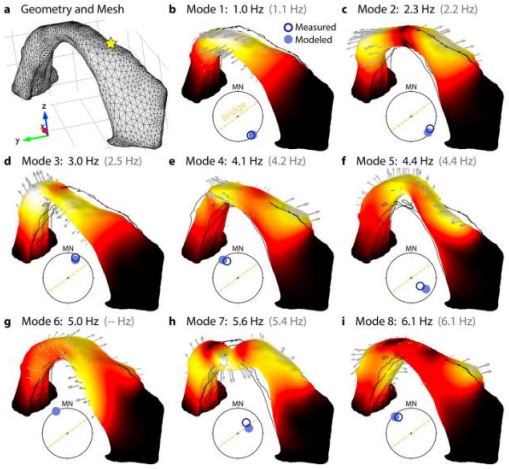
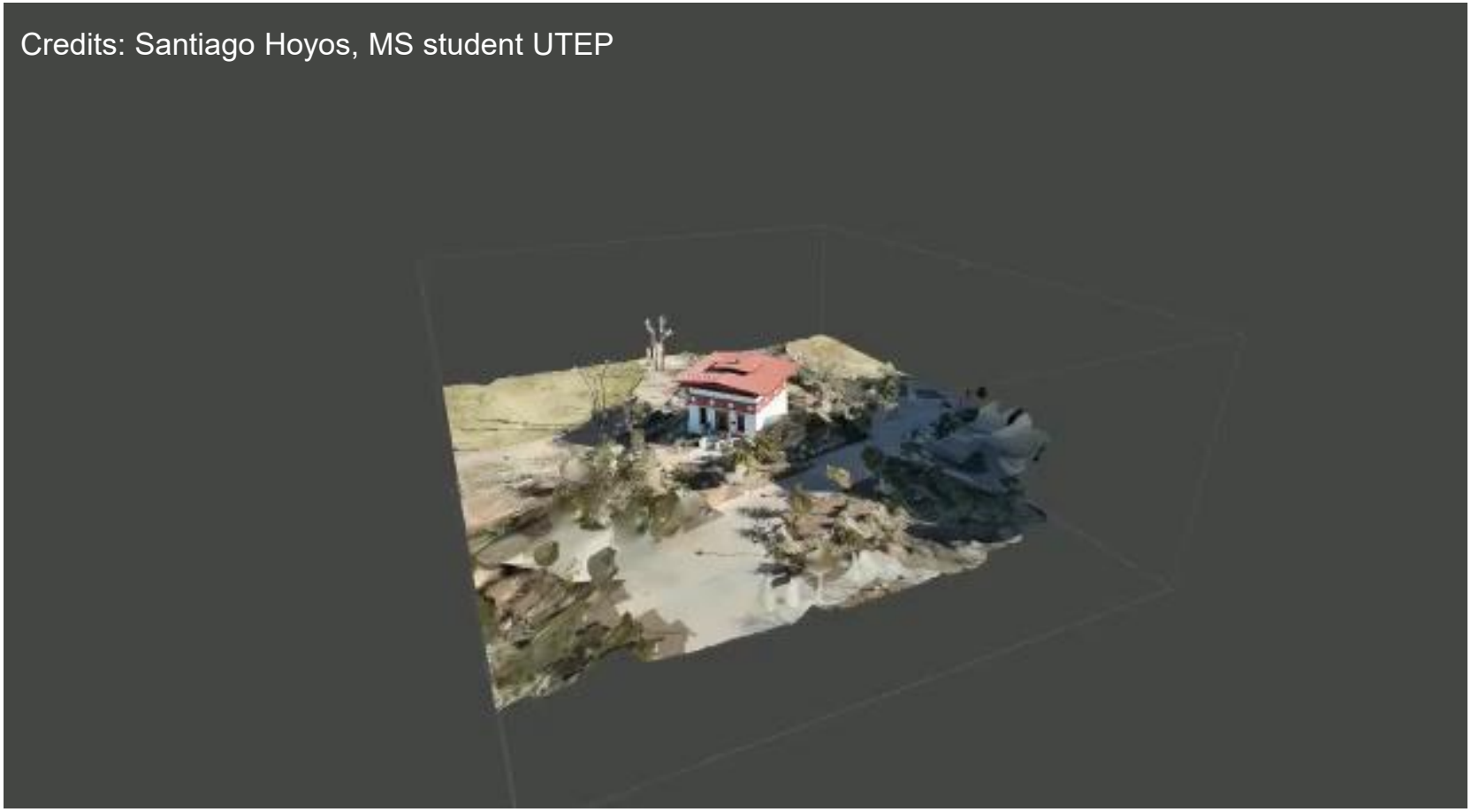


Figure 2. Numerical modal analysis of Rainbow Bridge. (a) Geometry and finite element mesh; *y* (magnetic north) and *z* (vertical) grid spacing = 20 m, *x* (east) grid spacing = 50 m. (b)–(i) First eight modes of vibration with accompanying eigenfrequency; measured values shown in parentheses. Color map, deformed body, and arrows illustrate deformation at zero phase (normalized relative scale for each mode), wireframe shows static form. Stereo plots compare measured (open circles) and modeled (filled) polarization vectors (see also Table 1); trend of Rainbow Bridge indicated by dashed line. MN = magnetic north.

Applications with Examples

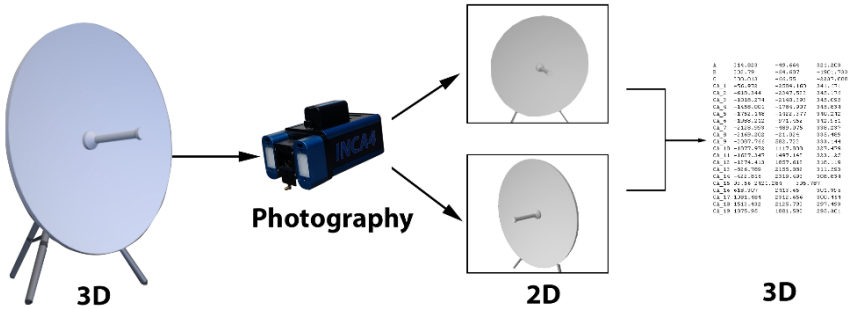
Credits: Santiago Hoyos, MS student UTEP

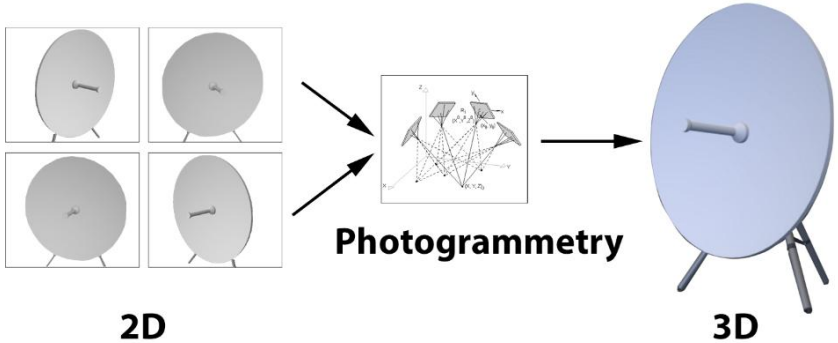


How to do ?

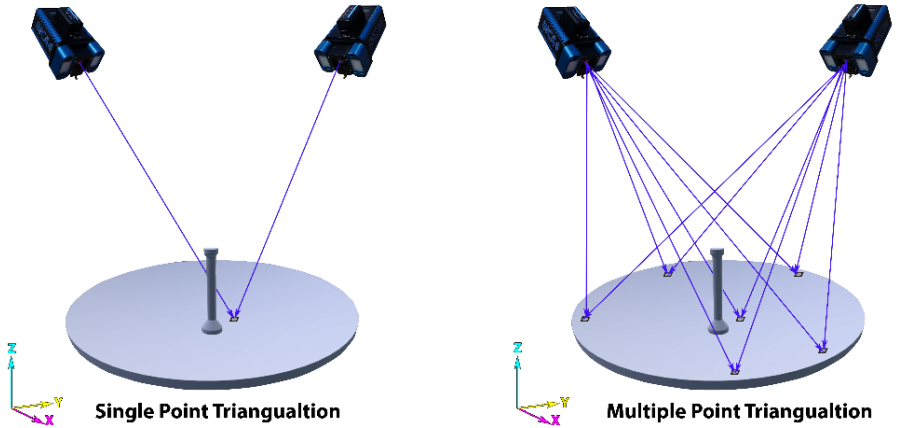
Focus on:
Focus Depth (F number)
Background and target exposure (Shutter Speed)

Have to have overlapping images
As a **minimum**, we require **two different photographs** to reconstruct 3D object





How to do ?



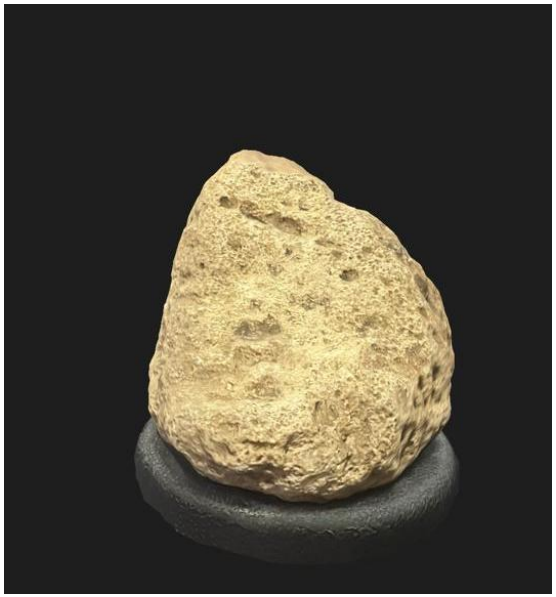
We can use post processing tools to derive the 3D object

Free Software
Meshroom, OpenMVG, VisualSFM, MicMac, COLMAP

Licensed
Agisoft Metashape

Our task

- Use our mobile phone (smart phones) to create 3D object
- https://www.kiriengine.com/
- Polycam



Reference

- DeWitt, B.A. and Wolf, P.R., 2000. *Elements of Photogrammetry (with Applications in GIS)*. McGraw-Hill Higher Education.
- Egodawela, S., Buddika, H.A.D.S., Harischandra, W.A.N.I., Mahmoodian, M. and Mathota, M.R.S.N.M., 2022, November. Conservation and Management of Ancient Stupa Using Digital Twins: A Case Study of Uav-Based 3D Photogrammetric Digitization of Rankoth Wehera Stupa. In *International Conference on Sustainable Built Environment* (pp. 493-508). Singapore: Springer Nature Singapore.
- Falkingham, P.L., Bates, K.T. and Farlow, J.O., 2014. Historical photogrammetry: Bird's Paluxy River dinosaur chase sequence digitally reconstructed as it was prior to excavation 70 years ago. PloS one, 9(4), p.e93247.
- Moore, J.R., Thorne, M.S., Koper, K.D., Wood, J.R., Goddard, K., Burlacu, R., Doyle, S., Stanfield, E. and White, B., 2016. Anthropogenic sources stimulate resonance of a natural rock bridge. Geophysical Research Letters, 43(18), pp.9669-9676